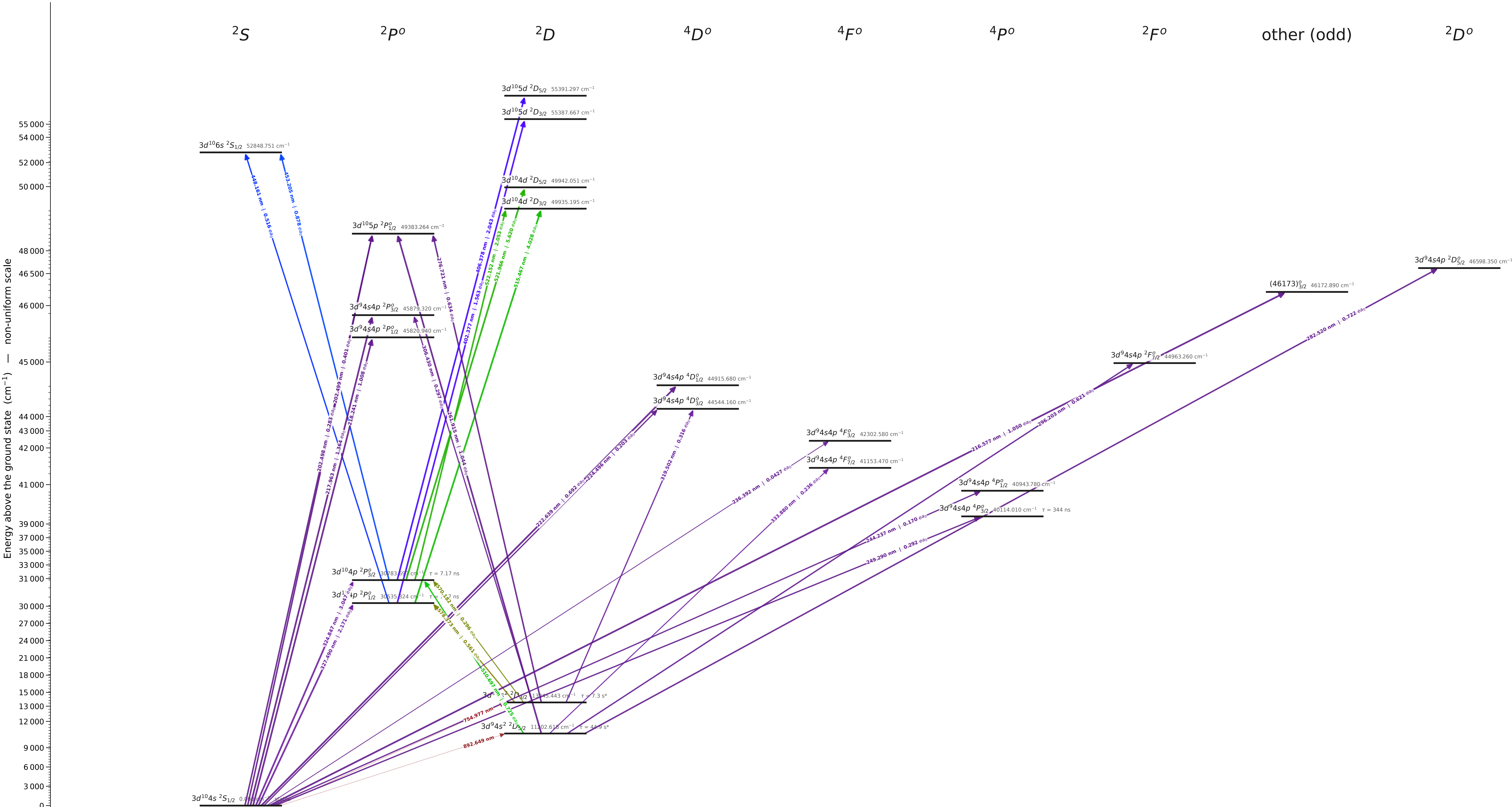
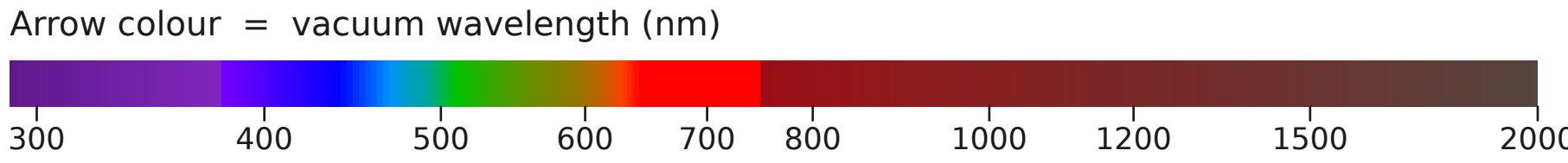


Copper

Cu (Cu I). Strongest classified transitions below 2 μm (NIST): 23 levels, 31 lines.

Cu^+ ionisation limit $62317.46 \text{ cm}^{-1} = 7.72638 \text{ eV}$ (160.469 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $\langle U || e r || U' \rangle$ in $e a_0$
 Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model calculation.
 Arrow width grows with the Einstein coefficient A . Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
 Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ \langle U er U\rangle $ (eao)	A (s ⁻¹)	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
$3d^{10}4s\ ^2S_{1/2} - 3d^94s4p\ ^4P_{3/2}$	249.2895	249.2143	1202.5878	0.292	2.790e+06	0.4627	96	D49.2 nm line used for resonance-ionisation spectroscopy...
$3d^{10}4s\ ^2S_{1/2} - 3d^{10}4p\ ^2P_{3/2}^o$	324.8473	324.7536	922.8720	3.047	1.372e+08	22.2	98.4	D2 resonance line 324.8 nm
$3d^{10}4s\ ^2S_{1/2} - 3d^{10}4p\ ^2P_{1/2}^o$	327.4896	327.3952	915.4260	2.171	1.359e+08	21.89	98.8	D1 resonance line 327.4 nm
$3d^{10}4s\ ^2S_{1/2} - 3d^94s^2\ ^2D_{3/2}$	754.9766	754.7688	307.0884	-	6.270e-07	2.18e-08	0	Second proposed clock transition
$3d^{10}4s\ ^2S_{1/2} - 3d^94s^2\ ^2D_{5/2}$	892.6485	892.4035	335.8460	-	2.230e-02	3.545e-09	100	E2 clock transition proposed for a Cu lattice clock

λ , ν from NIST level energies unless a measured frequency for this isotope is available. A : measured value or NIST $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

⁶³Cu hyperfine structure ($I = 3/2$)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)			
$3d^{10}4s^2 5s_{1/2}$	5866.915(5)	-	F=1: -7333.64	F=2: +4400.19		
$3d^94s^2 2D_{5/2}$	7491.00	186.000	F=1: -3802.57	F=2: -2453.18	F=3: -289.57	F=4: +2855.62
$3d^94s^2 2D_{3/2}$	1851.000	137.400	F=0: -6769.50	F=1: -5055.90	F=2: -1491.30	F=3: +4199.10
$3d^{10}4p^2 2P^o_{3/2}$	194.5(11)	-28.0(6)	F=0: -764.38	F=1: -541.88	F=2: -124.88	F=3: +430.62

Measured A, B constants; sources in the data file

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
$3d^{10}4s\ ^2S_{1/2} - 3d^94s4p\ ^4P^o_{3/2}$	249.290	63-65	1055.4(26)

$$\text{Shift} = \nu(\text{first isotope}) - \nu(\text{second isotope})$$